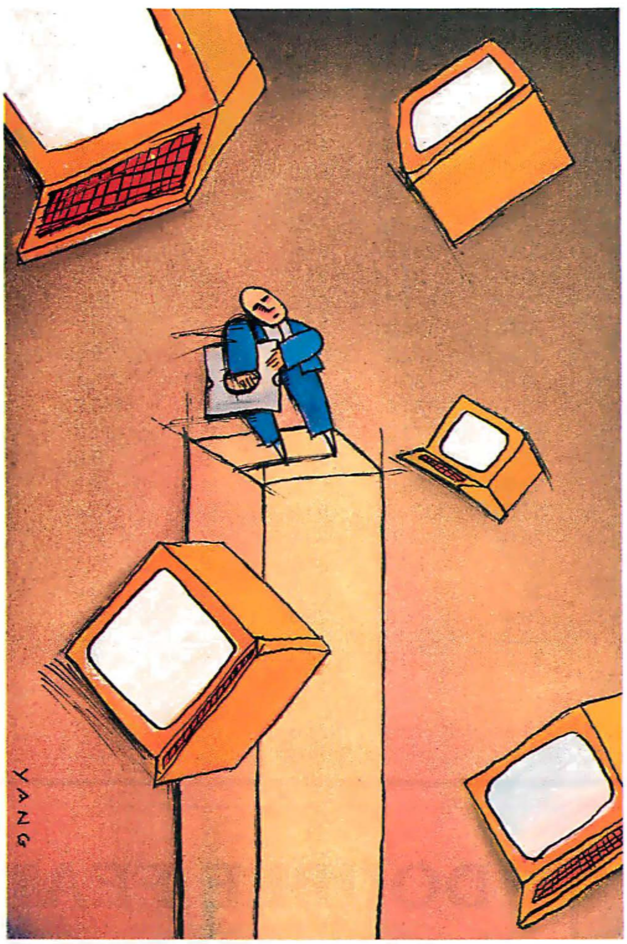
Multidimensional analysis: a new way to assess computer performance

Rating the IBM Compatibles

Robert G. Brookshire

A potential buyer of an IBM-compatible computer faces a bewildering variety of claims about features, capability, and performance. Despite their similarity to the IBM family of computers, these machines differ in a number of dimensions, making comparisons between them difficult. I will present a summary ranking of 35 IBM-compatible computers based on a multidimensional scaling analysis of seven of BYTE's standard measures of performance. You should not use this ranking as the sole basis for evaluating these computers, but it will help to simplify the complex task of comparing them.



Measures of Computer Performance

Since June 1984, BYTE has used a standard set of benchmark tests in its personal-computer reviews. Two tests, the Sieve and Calculation benchmarks, measure the CPU's speed. Two others measure the speed with which the computer writes and reads a 64K-byte file to and from disk. A fifth test compares the times required to copy a 40K-byte file to a floppy disk using the COPY command. The reviewer conducts each test three times and averages the results. I did not use the results of the Disk Copy test in this analysis, since this test is not applied to all machines. [Editor's note: *For more details on these benchmarks, see "Benchmarking the Clones" by Jon R. Edwards and Glenn Hartwig in BYTE's* Inside the IBM PCs, *Fall 1985.*]

Two additional benchmarks test spreadsheet performance (using Microsoft Multiplan). These tests measure the speed with which the computers load and recalculate a spreadsheet with 25 rows and 25 columns, in which each cell is 1.001 times the cell to its left. Table 1 shows the results of the seven benchmarks, as well as summary statistics for 35 personal computers that use the MS-DOS or PC-DOS operating systems.

I omitted the Hewlett-Packard 110 and the Stearns Desktop Computer because not all their benchmark results were published in BYTE. Results for the IBM PC, which are routinely reported in each review, are included. The model number and clock speed of the CPUs used in these computers (as reported in the reviews) is also in table 1.

The seven benchmark tests give seven separate measures of performance for personal computers. Although the result of each test is informative in itself, it is difficult to compare two or more computers that have seven sources of performance variation. BYTE does not provide a summary measure of performance based on these tests. It is possible, however, to compare and summarize the benchmark results for a large number of computers by using multidimensional scaling.

The Method

Multidimensional scaling is a mathematical technique for representing the configuration of variates in one-, two-, or higher-dimensional space (see references 1 and 2 for a thorough explanation of the method). It is applied in the social and behavioral sciences to reveal the structure that underlies relationships between objects, or to reduce the dimensionality of a set of variables. The data that is input to a multi-

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Table 1: *Benchmark test results and CPU characteristics for 35 personal computers. All times are in seconds.*

Computer	Disk Write	Disk Read	Sieve	Calculation	File Copy	Spreadsheet Load	Spreadsheet Recalculate	CPU Type	Clock Speed (MHz)
Epson Equity III	25.0	24.0	54.0	16.0	4.7	6.2	2.8	80286	8
Western AT	26.0	24.0	43.0	13.0	4.2	6.2	2.5	80286	8
Zenith Z-241	26.0	25.0	57.0	17.0	5.7	6.3	3.0	80286	6
Zenith Z-248	27.0	25.0	43.0	13.0	4.8	6.0	2.5	80286	8
IBM PC	56.0	46.0	191.0	69.0	5.8	8.1	11.0	8088	4.77
Epson Equity I	56.0	54.0	191.0	58.0	7.7	9.5	9.9	8088	4.77
Kaypro PC	55.0	30.0	186.0	57.0	8.2	8.6	10.0	8088	4.77
MPF-PC/700 D1	31.0	29.0	114.0	34.0	7.6	7.7	6.1	8088-2	8
Compaq Portable II	31.0	29.0	54.0	16.0	5.8	6.5	2.8	80286	8
Leading Edge Model D	56.0	30.0	190.0	58.0	8.2	8.1	10.0	8088	4.77
Xerox 6060	22.0	14.0	85.0	26.0	6.7	6.6	4.6	8086	N/A
NCR PC6	31.0	30.0	113.0	34.0	6.0	8.1	6.0	8088-2	8
Sperry PC/IT	9.0	8.8	44.0	14.0	.7	1.2	2.0	80286	7.16
ITT XTRA XP	10.0	10.0	60.0	20.0	4.8	2.9	3.2	80286	6
Conquest Turbo PC	30.0	30.0	130.0	40.0	9.5	9.4	7.5	8088-2	8
Deskpro 286	26.0	30.0	54.0	16.0	9.1	5.7	2.9	80286	8
Tele-286	25.0	24.0	54.0	16.0	3.5	4.9	2.8	80286	8
Executive Partner	31.0	29.0	90.0	28.0	9.1	6.1	5.1	8086-2	7.16
Kaypro 286i	24.0	23.0	73.0	22.0	8.2	7.3	4.1	80286	6
Canon A-200	57.0	29.0	132.0	41.0	13.0	8.0	7.4	8086	4.77
Color Fox	35.0	30.0	297.0	94.0	11.0	13.0	17.0	8088	3.6
AT&T PC 6300	32.0	30.0	87.0	27.0	10.0	7.0	4.9	8086-2	8
Data General/One	56.0	55.0	229.0	69.0	12.0	26.0	14.0	80C88	N/A
Sanyo MBC-775	30.0	29.0	154.0	35.0	8.0	7.7	6.3	8088-2	8
Ericsson	57.0	31.0	182.0	56.0	9.3	9.4	10.0	8088	4.77
Portable STM	31.0	30.0	80.0	24.0	5.7	5.1	3.7	80186	8
Kaypro 16	57.0	30.0	184.0	56.0	7.3	8.8	10.0	8088	N/A
Osborne 3	59.0	56.0	273.0	83.0	15.0	11.0	23.0	80C86	3.5
Tandy 1000	56.0	55.0	226.0	68.0	9.9	9.0	11.0	8088	4.77
TI Pro-Lite	34.0	33.0	155.0	51.0	11.0	7.2	9.1	80C88	5
Mindset	58.0	55.0	301.0	54.0	12.0	8.4	9.9	80186	6
Compaq Deskpro	30.0	29.0	93.0	29.0	7.4	6.1	5.1	8086	7.14
IBM PC AT	26.0	24.0	80.0	27.0	3.9	5.5	4.1	80286	6
ITT XTRA	33.0	32.0	185.0	56.0	8.8	10.0	10.0	8088	5
NEC APC III	30.3	29.4	86.0	28.9	6.5	8.4	6.7	8086-Z	8
Mean	36.5	31.2	130.6	39.0	7.8	7.9	7.2		
Standard Deviation	14.8	11.8	75.6	21.8	3.0	3.8	4.6		

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Table 2: *Multidimensional scaling results for 35 personal computers.*

Computer	Rank	Coordinate
Sperry PC/IT	1	1.8452
ITT XTRA XP	2	1.4024
Tele-286	3	1.0418
Western AT	4	1.0104
Zenith Z-248	5	.9886
Epson Equity III	6	.9317
IBM PC AT	7	.7919
Zenith Z-241	8	.7868
Xerox 6060	9	.7659
Compaq Portable II	10	.7341
Compaq Deskpro 286	11	.6769
Kaypro 286i	12	.6245
Portable STM	13	.6085
Compaq Deskpro	14	.4188
Panasonic Executive Partner	15	.3473
NEC APC III	16	.3368
AT&T PC 6300	17	.2956
NCR PC6	18	.2876
Multitech MPF-PC/700 D1	19	.2333
Sanyo MBC-775	20	.1258
Conquest Turbo PC	21	−.0368
TI Pro-Lite	22	−.2730
ITT XTRA	23	−.3715
Canon A-200	24	−.4747
Kaypro 16	25	−.5154
Kaypro PC	26	−.5270
Leading Edge Model D	27	−.5312
Ericsson	28	−.6030
IBM PC	29	−.8110
Epson Equity I	30	−.9019
Tandy 1000	31	−1.2126
Mindset	32	−1.4164
Color Fox	33	−1.6591
Osborne 3	34	−2.4205
Data General/One	35	−2.4999

Young's S-Stress formula: 1: 0.086
Kruskal's Stress formula: 1: 0.112
R squared: 0.967

dimensional scaling program is a measure of either similarity or dissimilarity between each pair of objects or variables that are to be scaled.

For this analysis, I calculated measures of dissimilarity between each pair of personal computers. I standardized each of the seven performance measures for each computer by converting it to a Z score, taking the difference between each test result and the average of that benchmark for all the machines, and dividing by the standard deviation. I then calculated a Euclidean distance measure for each pair of computers by taking the square root of the sum of the squared differences between the Z scores. Next, I used Kruskal's nonmetric multidimensional scaling method (see references 3 and 4) as implemented in the ALSCAL procedure of the SPSSx statistical package to reduce the resulting matrix of dissimilarities to one dimension.

Table 2 presents the results of the analysis. It reports each computer's coordinate on the dimension derived from the multidimensional scaling analysis, as well as the rank of the coordinate. The values for the measures of stress and R squared indicate a good fit of the model to the data; not much violence has been done to the data by reducing it to one dimension. Sperry's PC/IT is the best performer, while the Data General/One is the slowest overall.

The Results

The first dozen computers in table 2, with the exception of the Xerox 6060, are based on the Intel 80286 CPU. This processor is considerably faster than the Intel 8088 used in the IBM PC and most of the other computers in the analysis, and this accounts for the high performance rankings of these machines.

Within the 80286 machines, some of the factors that explain their relative rankings are the clock speed at which the CPU operates, the use of wait states by the

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CPU to synchronize timing with the memory chips or other devices, and the speed of the disk drives, all of which will affect the performance benchmarks. The Sperry PC/IT, for example, runs at a clock speed of 7.16 megahertz with no wait states and has a disk drive with a 30-millisecond average seek time. The IBM PC AT, in contrast, was benchmarked running at a 6-MHz clock speed with one wait state and using a hard disk drive with a 40-ms average access time, while the Zenith Z-248 used an 8-MHz clock with no wait states and a hard disk drive slightly faster than that of the PC AT. These factors, and probably others, have contributed to the large variation in the performance of the 11 IBM PC AT-compatible machines analyzed here.

Of the next five ranking computers, four use variants of Intel's 8086 CPU, while the Portable STM uses the 80186. Since the 80186 can be considered an intermediate chip between the 8086 and the 80286, the placement of the STM in the ranking is not surprising. It appears, based on its coordinate, to be more similar in performance to the 80286 machines than the 8086-based computers.

The Xerox 6060, which uses the 8086, outperforms all the other 8086 computers, as well as several of the 80286-based systems. This is due to the machine's rapid execution of the disk-performance benchmarks. Among the other 8086 machines, it is interesting that the AT&T and NEC computers, although operating at a clock speed of 8 MHz, are slower in overall performance than the Compaq Deskpro and Panasonic systems, which run at slower clock speeds. The machines with faster clock speeds did execute some benchmark tests more rapidly than the Compaq and Panasonic systems but were slower on other measures. This illustrates the usefulness of a multivariate analysis, which can summarize all seven performance measures simultaneously.

On the other hand, the coordinates on the derived dimension indicate no tremendous difference in the performance of these four 8086 machines (AT&T, NEC, Compaq Deskpro, and Panasonic). The computers ranking from 18 to 21 use Intel's 8088-2 processor, which can be set at either a 4.77- or 8-MHz clock speed. All benchmark results used in this analysis were those for the faster clock speed. Since these four machines use the same processor and clock speed, one might expect to find similar coordinates on the derived dimension due to similar results on the benchmark tests.

This is not the case. Considerable variation in the coordinates reflects variation on the underlying measures. As table 1 shows, these machines differ most on the Sieve benchmark, which measures processor speed. Even for computers with the same processor and clock speed, performance can differ significantly.

Nine of the remaining 14 computers use Intel's 8088 processor, including the IBM PC. As table 2 indicates, the IBM PC has relatively poor performance when compared with the other machines in this analysis. Even among the nine machines using the 8088, the IBM ranks sixth in speed. On the other hand, the IBM PC is the oldest machine on the list, and there are few dramatic differences between it and the other machines of its type.

The two lowest-ranking machines, the Data General/One and the Osborne 3, and the higher-ranking TI Pro-Lite, are "laptops," small and light enough to be held in your lap. Most laptop computers use variants of the Intel 80C88 processor, a CMOS version of the 8088 that uses significantly less electrical power than standard microprocessors, making it more suitable for battery-powered computers.

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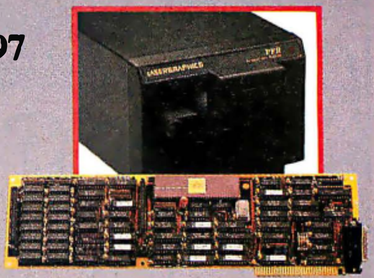

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Lite performs slightly better than its 8088-based cousins, but the Data General and Osborne computers are extraordinarily slow. The other two slow systems, the Color Fox and the Mindset, have enhancements for graphics applications that apparently degrade performance.

Other Considerations

You should take many factors into consideration when purchasing a personal computer, including price, warranty terms, and availability of repair service. Also, you must consider software compatibility, accessibility of training and consultation, and performance.

The performance dimension has several sources of variation, however. These include the type of CPU used in the computer, the CPU's clock speed, the use of wait states, and the speed of the disk drives. All these sources of variation interact to affect performance in ways difficult to evaluate.

Using standard benchmarking programs is one approach to comparing the performance of computers. However, when you use several benchmarks, the effect can be to provide additional sources of variability at a higher level of abstraction. I've used multidimensional scaling to summarize seven benchmark tests and yield a rank order of 35 personal computers based on performance.

The ranking shows that, in general, computers using the Intel 80286 processor are the fastest, followed by those using the 8086, the 8088-2, and the 8088, with laptop computers using the 80C88 processor and its variants generally the slowest. Within each of these broad categories is considerable diversity. Some of it can be accounted for by CPU clock speed, wait states, or disk drive speed, but some differences in performance remain even among machines with almost identical designs.

The lesson to be learned from this analysis is that, in microcomputer performance, the proof is in the using. Those who are concerned about performance should measure the speed of the computers in performing tasks for which they are likely to use the machines. These performance tests are critical even for machines with supposedly equivalent architectures and components. A series of benchmarking programs can capture important differences in performance that design specifications do not reveal.

Finally, a few words of caution about interpreting this performance ranking. Although the multidimensional-scaling-analysis results in this article are valid for the machines analyzed, they could vary significantly if other computers, particularly computers differing substantially in performance from those examined here, were introduced into the analysis. Adding a new machine to the ranking would require the calculation of new dissimilarities and a new analysis of the entire set of data. It is not possible to infer from this particular ranking where a machine not included in this analysis would fall. ■

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